

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Video Projector
Manufacturer: Christie
Firmware Version: N/A
Model(s): LX120

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.00.0000-b022	2.0.0

Version History

Module Version	Date	Notes
1_0_0_0	10/16/2018	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='LX120')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='LX120')</code>

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AspectRatio ¹	Value 'Normal'	Value 'Wide'	Value 'True'
# AspectRatio example InterfaceName.Set('AspectRatio', 'Normal')			
Command AudioMute	Value 'On'	Value 'Off'	
# AudioMute example InterfaceName.Set('AudioMute', 'On')			
Command AutoImage	Value None		
# AutoImage example InterfaceName.Set('AutoImage', None)			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Set('Freeze', 'On')			
Command Input ²	Value 'Digital' 'S-Video' 'YCbCr'	Value 'Analog' 'Y/C' 'HD-SDI'	Value 'Video' 'YPbPr'
Qualifier Key 'Input'	Qualifier Value '1' – '4'		
# Input example InterfaceName.Set('Input', 'Digital', {'Input': '1'})			
Command LampMode	Value 'Full'	Value 'Half'	
# LampMode example InterfaceName.Set('LampMode', 'Full')			
Command MenuNavigation	Value 'Up' 'Right' 'Menu Off'	Value 'Down' 'Select'	Value 'Left' 'Menu On'
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Up')			

**Global Scripter Module
Communication Sheet**

Command Power ²	Value 'On'	Value 'Off'
# Power example InterfaceName.Set('Power', 'On')		
Command VideoMute	Value 'On'	Value 'Off'
# VideoMute example InterfaceName.Set('VideoMute', 'On')		
Command Volume	Value 0 to 63 in steps of 1	
# Volume example InterfaceName.Set('Volume', 63)		

¹ A 1000ms delay is needed between commands.² A 5000ms delay is needed between commands.

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus and DeviceStatus do not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command AspectRatio	Value 'Normal'	Value 'Wide'	
# AspectRatio examples InterfaceName.Update('AspectRatio') Value = InterfaceName.ReadStatus('AspectRatio') InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)			
Command AudioMute	Value 'On'	Value 'Off'	
# AudioMute examples InterfaceName.Update('AudioMute') Value = InterfaceName.ReadStatus('AudioMute') InterfaceName.SubscribeStatus('AudioMute', None, FeedbackHandler)			
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
# ConnectionStatus examples Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command DeviceStatus	Value 'Normal'	Value 'Lamps are off due to Power Management'	Value 'Power Failure'
	'Temperature warning and cooling down'	'Temperature warning and power is ON'	'Temperature warning occurred and system has recovered'
	'No key input'	'Lamp is OFF and Temperature warning by Power Management'	'Lamp off & Cooling Down'
	'Standby after lamp off & cooling down'		

Global Scripter Module Communication Sheet

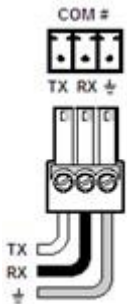
<pre># DeviceStatus examples Value = InterfaceName.ReadStatus('DeviceStatus') InterfaceName.SubscribeStatus('DeviceStatus', None, FeedbackHandler)</pre>			
Command Freeze	Value 'On'	Value 'Off'	
<pre># Freeze examples InterfaceName.Update('Freeze') Value = InterfaceName.ReadStatus('Freeze') InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)</pre>			
Command Input	Value 'Digital' 'S-Video' 'YCbCr' 'Blank'	Value 'Analog' 'Y/C' 'HD-SDI'	Value 'Video' 'YPbPr' 'No Card'
Qualifier Key 'Input'	Qualifier Value '1' – '4'		
<pre># Input examples InterfaceName.Update('Input', {'Input': '1'}) Value = InterfaceName.ReadStatus('Input', {'Input': '1'}) InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)</pre>			
Command LampMode	Value 'Full'	Value 'Half'	
<pre># LampMode examples InterfaceName.Update('LampMode') Value = InterfaceName.ReadStatus('LampMode') InterfaceName.SubscribeStatus('LampMode', None, FeedbackHandler)</pre>			
Command LampUsage	Value Hours		
Qualifier Key 'Lamp'	Qualifier Value '1' – '4'		
<pre># LampUsage examples InterfaceName.Update('LampUsage', {'Lamp': '1'}) Value = InterfaceName.ReadStatus('LampUsage', {'Lamp': '1'}) InterfaceName.SubscribeStatus('LampUsage', None, FeedbackHandler)</pre>			
Command Power	Value 'On' 'Cooling Down'	Value 'Off'	Value 'Warming Up'
<pre># Power examples InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)</pre>			
Command VideoMute	Value 'On'	Value 'Off'	
<pre># VideoMute examples</pre>			

<pre>InterfaceName.Update('VideoMute') Value = InterfaceName.ReadStatus('VideoMute') InterfaceName.SubscribeStatus('VideoMute', None, FeedbackHandler)</pre>	
Command Volume	Value 0 to 63 in steps of 1
<pre># Volume examples InterfaceName.Update('Volume') Value = InterfaceName.ReadStatus('Volume') InterfaceName.SubscribeStatus('Volume', None, FeedbackHandler)</pre>	

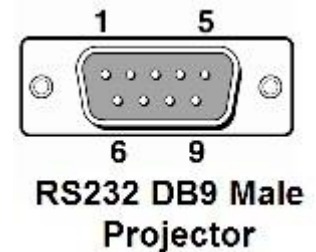
Cable and Adapter Requirements

Captive Screw to Female DB9 RS-232 Serial Cable

Notes for the Device

Serial communication**Port Type:** RS-232**Baud Rate:** 19200**Data Bits:** 8**Parity:** None**Stop Bits:** One**Flow Control:** None**Pin Assignments Diagram**

Signal	Main Cable	Pin	Signal
TxD		2	TxD
RxD		3	RxD
GND		5	GND



Appendix A. Set Commands

Aspect Ratio Normal	CF SCREEN NORMAL\x0D
Aspect Ratio True	CF SCREEN TRUE\x0D
Aspect Ratio Wide	CF SCREEN WIDE\x0D
Audio Mute Off	C0C\x0D
Audio Mute On	C0B\x0D
Auto Image None	C89\x0D
Freeze Off	C44\x0D
Freeze On	C43\x0D
Input Analog Input 1	CF INPUT1 ANALOG\x0D
Input Analog Input 4	CF INPUT4 ANALOG\x0D
Input Digital Input 1	CF INPUT1 DIGITAL\x0D
Input Digital Input 4	CF INPUT4 DIGITAL\x0D
Input HD-SDI Input 1	CF INPUT1 HD-SDI\x0D
Input HD-SDI Input 4	CF INPUT4 HD-SDI\x0D
Input S-Video Input 1	CF INPUT1 S-VIDEO\x0D
Input S-Video Input 4	CF INPUT4 S-VIDEO\x0D
Input Video Input 1	CF INPUT1 VIDEO\x0D
Input Video Input 4	CF INPUT4 VIDEO\x0D
Input Y/C Input 1	CF INPUT1 YC\x0D
Input Y/C Input 4	CF INPUT4 YC\x0D
Input YCbCr Input 1	CF INPUT1 YCBCR\x0D
Input YCbCr Input 4	CF INPUT4 YCBCR\x0D
Input YPbPr Input 1	CF INPUT1 YPBPR\x0D
Input YPbPr Input 4	CF INPUT4 YPBPR\x0D
Lamp Mode Full	CF LAMPMODE FULL\x0D
Lamp Mode Half	CF LAMPMODE HALF\x0D
Menu Navigation Down	C3D\x0D
Menu Navigation Left	C3B\x0D
Menu Navigation Menu Off	C1D\x0D
Menu Navigation Menu On	C1C\x0D
Menu Navigation Right	C3C\x0D
Menu Navigation Select	C3F\x0D
Menu Navigation Up	C3C\x0D
Power Off	C01\x0D
Power On	C00\x0D
Video Mute Off	C0E\x0D

**Global Scripter Module
Communication Sheet**

Video Mute On	C0D\x0D
Volume 0	CF VOLUME 000\x0D
Volume 63	CF VOLUME 063\x0D

Appendix B. Update Commands

Aspect Ratio	CR SCREEN\x0D
Audio Mute	CR MUTE\x0D
Freeze	CR FREEZE\x0D
Input Input 1	CR SRCINP1\x0D
Input Input 4	CR SRCINP4\x0D
Lamp Mode	CR LAMPMODE\x0D
Lamp Usage Lamp 1	CR3\x0D
Lamp Usage Lamp 4	CR3\x0D
Power	CR0\x0D
Video Mute	CR VMUTE\x0D
Volume	CR VOLUME\x0D